

US Patent & Trademark Office

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United States Patent Application Publication

20250281121

Kind Code

A1

Publication Date

September 11, 2025

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BREAST IMPLANT AND APPARATUS FOR SENSING ABNORMALITY OF THE SAME

Abstract

This application relates to a breast implant and an apparatus for sensing an abnormality of the breast implant. In one aspect, the breast implant includes a shell including an injection hole and forming an outer cover of the breast implant and a filler of a gel state injected into the shell through the injection hole. The breast implant may also include a first patch configured to seal the injection hole, the first patch including a first sensor module. The breast implant may further include a second patch configured to be attached on at least one of a second position facing the first position and a third position to which a pressure due to a gravity is applied when the breast implant is inserted in a breast and a body is kept upright without an external pressure applied, the second patch including a second sensor module.

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Appl. No.: 19/215800

Filed: May 22, 2025

Foreign Application Priority Data

KR 10-2021-0022995

Feb. 21, 2021

KR 10-2021-0050179

Apr. 17, 2021

Related U.S. Application Data

parent US continuation 17671357 20220214 parent-grant-document US 12329541 child US 19215800

Publication Classification

Int. Cl.: A61B5/00 (20060101); A61F2/12 (20060101); A61L27/52 (20060101)

U.S. Cl.:

CPC A61B5/686 (20130101); A61B5/0022 (20130101); A61B5/746 (20130101); A61F2/12 (20130101); A61L27/52 (20130101); A61B2562/0247 (20130101); A61F2250/0003 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/KR2021/013546, filed Oct. 3, 2021, which is based upon and claims the benefit of priority from Korean Patent Application No. 10-2021-0022995 filed Feb. 21, 2021 and Korean Patent Application No. 10-2021-0050179 filed Apr. 17, 2021, which is now Korean Patent No. 10-2324101, the contents of which are incorporated herein by reference in their entirety.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a breast implant and an apparatus for sensing an abnormality of the breast implant.

Description of the Related Technology

[0003] A breast with beauty and volume is the shared wishes of most women. The breast implant is one of the easiest and most effective methods to realize the wishes of such women.

[0004] The breast reconstruction employing the breast implant is a method of securing a space for inserting the breast implant in a breast and increasing the volume of the breast by inserting the breast implant in the secured space.

SUMMARY

[0005] According to some embodiments of the present disclosure, a breast implant includes a shell including an injection hole and forming an outer cover of the breast implant, a filler of a gel state injected into the shell through the injection hole, a first patch configured to seal the injection hole, the first patch including a first sensor module, and a second patch configured to be attached on at least one of a second position facing the first position and a third position to which a pressure due to a gravity is applied when the breast implant is inserted in a breast and a body is kept upright without an external pressure applied, the second patch including a second sensor module.

[0006] According to some embodiments of the present disclosure, an apparatus for sensing an abnormality of a breast implant includes a wireless power transmitting unit configured to transmit a power in a wireless manner, an abnormality sensing unit configured to receive a pressure signal from at least one of the first sensor module and the second sensor module of the breast implant according to some embodiments of the present disclosure and to sense an abnormality of the breast implant based on the pressure signal, and an alarm generating unit configured to, upon the abnormality sensing unit sensing the abnormality of the breast implant, generate an alarm.

[0007] The above and other objects, features, advantages and technical and industrial significance of this invention will be better understood by reading the following detailed description of presently preferred embodiments of the invention, when considered in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic diagram for explaining pressures applied to a breast implant inserted in a breast in some embodiments of the present disclosure.

[0009] FIG. 2 is a schematic diagram of a breast implant according to some embodiments of the present disclosure.

[0010] FIGS. 3 and 4 are schematic diagrams of a patch including a sensor module according to some embodiments of the present disclosure.

[0011] FIG. 5 is an exploded view of a sensor module according to some embodiments of the present disclosure.

[0012] FIGS. 6 and 7 are schematic diagrams of an apparatus for sensing an abnormality of a breast implant according to some embodiments of the present disclosure.

[0013] FIG. 8 is a schematic diagram showing an example of attaching a patch including a sensor module according to some embodiments of the present disclosure.

[0014] FIG. 9 is a functional block diagram of an apparatus for sensing an abnormality of a breast implant according to some embodiments of the present disclosure.

[0015] FIGS. 10 to 13 are flowcharts for explaining a method of manufacturing a breast implant according to some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0016] Breast implants are generally manufactured using a cohesive gel that is a silicone having a high cohesive force. The cohesive gel has an advantage in that an outer cover (shell) is strong and a shape is naturally formed after a surgery, remedying disadvantages of a conventional silicone gel (see, for example, Korean Pat. No. 10-1235284, Korean Pat. No. 10-1966979, and Japanese Pat. Laid-Open No. 2010-534551).

[0017] Popular side effects seen after a surgery for reconstructing a breast with a breast implant include rupture and capsular contracture.

[0018] When a rupture is generated and a silicone component is remained in the body for a long time, the silicone component may permeate through the breast tissue, possibly causing a breast feeding hard and a difficulty in a breast cancer examination. Sometimes the silicone penetrates into a lymph node due to the rupture of the breast implant, and in this case, the penetrated silicone may be incompletely removed and a lymphedema after removing the silicone or a spreading of the silicone to the whole body may occur.

[0019] The capsular contracture is a phenomenon generated from thickening of a capsule newly formed surrounding a breast implant, which is generated within the first two years after a surgery, which is one of the most popular side effects.

[0020] A capsular contracture causes deformation and pain, and in a worse case, a reoperation is needed to remove the breast implant and to remove the capsule.

[0021] In order to examine the two representative side effects after inserting the breast implant, a patient needs to visit a hospital to consult a doctor and to carry out an image medical inspection such as an ultrasound and an MRI.

[0022] Even without a particular symptom, it is recommended to carry out an ultrasound examination every year after inserting the breast implant.

[0023] Exemplary embodiments of the present disclosure are described in detail below with reference to the accompanying drawings.

[0024] It is an object of the present disclosure to provide a breast implant with which an abnormality of the breast implant can be effectively sensed after inserting the breast implant in a breast.

[0025] It is an object of the present disclosure to provide an apparatus for sensing an abnormality

of a breast implant, which can effectively sense an abnormality of the breast implant after inserting the breast implant in a breast.

[0026] The challenges to be addressed by the present disclosure are not limited to those mentioned above, and other unmentioned problems can be clearly understood by those skilled in the art from the following description.

[0027] A breast implant according to some embodiments of the present disclosure is manufactured with, for example, a silicone material and is used for a plastic surgery of a breast, a hip, or the like, which is inserted in a body and forms and maintains a consistent shape in the body.

[0028] Although a breast implant for reconstructing a breast is explained as an example in the present specification, the breast implant and the apparatus for sensing an abnormality of a breast implant according to some embodiments of the present disclosure can be applied to most implants including a shell and a gel-type filler and being inserted in a body for a specific purpose.

[0029] FIG. 1 is a schematic diagram for explaining pressures applied to a breast implant inserted in a breast in some embodiments of the present disclosure.

[0030] A breast implant for reconstructing a breast is explained as an example in FIG. 1. A plastic surgery of a breast employing a breast implant is a method of securing a space for inserting the breast implant in a breast and increasing the volume of the breast by inserting the breast implant in the secured space.

[0031] As shown in FIG. 1, when a breast implant **110** is inserted in a breast **100**, in a normal state in which no external pressure is applied on the breast **100**, a pressure A is applied on the breast implant **110** from outside a body, a pressure B is applied on the breast implant **110** from or against inside the body, and pressure C is applied on the breast implant **110** due to the gravity.

[0032] Typically, a silicone gel is inserted in the shell of the breast implant **110**. The breast implant **110** is sealed by a patch at a position on the bottom of the breast implant **110** (near a position indicated by B) in a manner that consistent tension is maintained. However, when a part of the breast implant **110** is ruptured or a capsular contracture is generated at a part of the breast implant **110**, at least one of the pressures A, B, and C is changed.

[0033] In the case of the cohesive-gel implant, which is widely used in these days, even if a part of the breast implant **110** is ruptured after inserting the breast implant **110** in the breast **100**, a patient can hardly feel an abnormality, which is referred to as a silent rupture. That is, the cohesive gel keeps the gel state even when the breast implant **110** is ruptured.

[0034] An image inspection by a magnetic resonance imaging (MRI) is so far one of the most precise methods to inspect a rupture of the breast implant **110**. However, the MRI itself still shows a false positive and a false negative and can be a burden to a patient in terms of the cost.

[0035] When a foreign material comes into a body, a capsule is formed around the foreign material. Such capsule excessively formed and hardened causes a capsular contracture.

[0036] When a breast implant is inserted in a breast, the blood platelet in the blood is activated and a material called the transforming growth factor β (TGF- β) is secreted. The TGF- β takes a role of gathering monocytes around the breast implant where an inflammation is generated. The monocyte is a sort of the white blood cell, which develops into a macrophage near the inflammation and secretes the TGF- β again. This results in a fibroblast near the inflammation and a collagen synthesized from the fibroblast generates the capsular contracture.

[0037] FIG. 2 is a schematic diagram of the breast implant **110** according to some embodiments of the present disclosure.

[0038] As shown in FIG. 2, the breast implant **110** according to some embodiments of the present disclosure includes a shell **111** forming an outer cover of teardrop type or round type, a silicone gel **112** as a filler injected into the shell **111**, and a patch **200** for sealing an injection hole **113** for injecting the silicone gel **112**.

[0039] The patch **200** according to some embodiments of the present disclosure includes a sensor module **300** including a pressure sensor for sensing a pressure applied to the breast implant **110**.

[0040] FIG. 3 is a schematic diagram of the patch **200** including a sensor module **300** according to some embodiments of the present disclosure. FIG. 4 is a schematic diagram of a patch **400** including the sensor module **300** according to some embodiments of the present disclosure.

[0041] The patch **200** shown in FIG. 3 includes a first barrier (inner barrier) **211** forming a first side facing inside the breast implant **110** upon being attached on the breast implant **110** with the filler of the gel state injected, a second barrier (outer barrier) **212** facing the first barrier **211** and forming a second side opposite to the first side. The sensor module **300** is disposed on the first side of the first barrier **211**.

[0042] In the patch **400** shown in FIG. 4, the sensor module **300** is disposed between the first barrier **211** and the second barrier **212**.

[0043] The patch **200**, **400** is a unit member formed by the first barrier **211** and the second barrier **212** making contact with each other. The first barrier **211** forms the inner side of the patch **200**, **400**, and the second barrier **212** forms the outer side of the patch **200**, **400**.

[0044] Therefore, the patch **200** shown in FIG. 3 can be manufactured by forming a unit member with the first barrier **211** and the second barrier **212** having contact with each other and then attaching the sensor module **300** on a part of the outer surface of the first barrier **211**.

[0045] The patch **400** can be manufactured by sandwiching the sensor module **300** between the first barrier **211** and the second barrier **212**.

[0046] FIG. 5 is an exploded view of the sensor module **300** according to some embodiments of the present disclosure.

[0047] As shown in FIG. 5, the sensor module **300** is formed as a film including two protective films **310**, **330** and a substrate **320** between the two protective films **310**, **330**. On the substrate **320**, at least one pressure sensor **323** for sensing a pressure applied on the breast implant and outputting a pressure signal, a controller **322** for transmitting the pressure signal from the pressure sensor **323** to the outside, and a wireless power receiving unit **321** for receiving a power from the outside and providing the power to the controller **322**.

[0048] The pressure sensor **323** is an element that, when a predetermined pressure or force is applied, generates a signal proportional to the pressure or the force applied. The pressure sensor **323** includes a diaphragm sensor that curves according to the pressure or the force applied, a force sensing resistor (FSR) including two conductive plate and a conductive polymer between the two conductive plate, and a piezoelectric element that generates a voltage when a pressure is applied.

[0049] Although a tension sensor that senses a tension of the shell of the breast implant can be used instead of the pressure sensor, as a change of the tension is also a physical amount proportional to a change of the pressure due to the rupture or the capsular contracture of the breast implant, the tension sensor is considered as a type of the pressure sensor in the present specification.

[0050] In some embodiments of the present disclosure, the sensor module **300** further includes a battery **324** for storing the power received by the wireless power receiving unit **321** in a wireless manner.

[0051] That is, the sensor module **300** according to some embodiments of the present disclosure can operate either by receiving the power in a wireless manner only without a battery or by charging a battery with the power received in a wireless manner.

[0052] FIGS. 6 and 7 are schematic diagrams of an apparatus for sensing an abnormality of a breast implant according to some embodiments of the present disclosure.

[0053] The breast implant according to some embodiments of the present disclosure includes, as shown in FIG. 2, the shell **111** of the silicone material, the silicone gel **112** filled in the shell **111**, and at least one patch **200**, **400**.

[0054] An apparatus **600** for sensing an abnormality of a breast implant shown in FIG. 6 transmits the power to the sensor module **300** of the breast implant, receives the pressure signal from the sensor module **300**, determines the abnormality of the breast implant based on the pressure signal, and outputs a result of determination upon determining the abnormality.

[0055] The apparatus **600** includes a computing device including a computer and a digital pad such as a personal digital assistant (PDA) and a smart phone, having functions of the wireless communication and the wireless power transmission.

[0056] An apparatus **700** for sensing an abnormality of a breast implant shown in FIG. 7 has a form of a smart watch that is wearable. The apparatus **700** transmits the power to the sensor module **300** of the breast implant, receives the pressure signal from the sensor module **300**, determines the abnormality of the breast implant based on the pressure signal, and outputs a result of determination upon determining the abnormality.

[0057] In some embodiments of the present disclosure, the apparatus **600**, **700** stores, in a memory (not shown), an initial pressure sensed by the sensor module **300** when the breast implant **110** is inserted in the body, compares the pressure indicated by the pressure signal received from the sensor module **300** with the initial pressure, and senses the abnormality of the breast implant **110** based on a result of comparison.

[0058] That is, when there is a change in the pressure due to the rupture or the capsular contracture, a user is notified of the rupture or the capsular contracture of the breast implant according to the change of the pressure, to enable the user to take a proper action at the right time.

[0059] This can help the user to remove an anxiety about maintaining the breast with safety for a long time after inserting the breast implant in the breast, and at the same time, to relieve an inconvenience of taking an ultrasound inspection or an MRI inspection on a regular basis.

[0060] FIG. 8 is a schematic diagram showing an example of attaching a patch **200** and a patch **400** each including a sensor module according to some embodiments of the present disclosure.

[0061] As shown in FIG. 1, when a breast implant is inserted in a breast, in a normal state in which no external pressure is applied on the breast, a pressure A is applied on the breast implant **110** from outside a body, a pressure B is applied on the breast implant **110** from or against inside the body, and pressure C is applied on the breast implant **110** due to the gravity.

[0062] Therefore, by disposing the patch **200**, **400** on at least one of positions A, B, and C, it is possible to sense the change of the pressure when there is an abnormality in the breast implant and to notify the user of the abnormality.

[0063] As shown in FIG. 8, the patch **200** including the sensor module **300** on the surface of the first barrier **211** or the patch **400** including the sensor module **300** between the first barrier **211** and the second barrier **212** can be attached on the position B, and the patch **400** can be attached on the positions A and C not to allow the sensor module **300** to make direct contact with the skin tissue.

[0064] As shown in FIGS. 1 and 8, the position B may not show a large change of the pressure due to an abnormality of the breast implant as a fixed wall is formed against the body and the sensor module **300** of the patch **200** disposed at this position senses the pressure exerted on a large area forming the bottom of the breast implant **110**.

[0065] On the other hand, in the case of the position A or the position C, as a fluidity is relatively large compared to the position B, the change of the pressure due to the abnormality of the breast implant can be relatively large.

[0066] When an abnormality of the breast implant is sensed and, for example, it is determined that a rupture of the breast implant is sensed, the breast implant inserted in the breast is removed and a new breast implant is inserted in the breast again. Therefore, an error in the determination of the abnormality can cause a considerably negative consequence.

[0067] In order to prevent such errors in the determination of the abnormality, in some embodiments of the present disclosure, a first patch (patch **200**) and a second patch (patch **400**) are attached on at least two positions among the position B (first position), the position A (second position), and the position C (third position) to minimize the error in the determination of the abnormality due to the relative magnitude of the change in the pressure.

[0068] In some embodiments of the present disclosure, the second patch (patch **400**) attached on the position A (second position) and the position C (third position) is smaller than the first patch

(patch **200**) attached on the position B (first position) in diameter.

[0069] This is because, as shown in FIG. **8**, the position A (second position) and the position C (third position) have smaller areas flat enough to dispose the patch compared to the position B (first position).

[0070] For the same reason, the second patch (patch **400**) attached on the position C (third position) can have a diameter smaller than that of the second patch (patch **400**) attached on the position A (second position).

[0071] In some embodiments of the present disclosure, the apparatus **600, 700** stores the initial pressure for each of a plurality of positions, for example, the positions A, B, and C shown in FIG. **1** in the memory.

[0072] The apparatus **600, 700** may output an abnormality of the breast implant when a difference between the pressure indicated by at least one of the pressure signals received from the sensor modules **300** disposed at the plurality of positions and the initial pressure stored in the memory in association with the corresponding position exceeds a predetermined threshold.

[0073] In some embodiments of the present disclosure, the apparatus **600, 700** sets a threshold for determining an abnormality in a pressure signal received from one sensor module **300** larger than thresholds for determining an abnormality in pressure signals received from the other sensor modules **300**. By configuring the apparatus **600, 700** in this manner, it is possible to determine the abnormality with the pressure signal from the one sensor module when the degree of the abnormality is considerable and to determine no abnormality when the degree of the abnormality is trivial and an abnormality in one sensor module is permissible, which can avoid an unnecessary alarming of the abnormality.

[0074] In some embodiments of the present disclosure, the apparatus **600, 700** sets different thresholds for a plurality of sensor modules from each other to determine the abnormality. For example, the apparatus **600, 700** sets the threshold for the sensor module on the bottom of the breast implant (position C in FIG. **1**) larger or smaller than the thresholds for the sensor modules on the other positions. By configuring the apparatus **600, 700** in this manner, it is possible to make a proper determination considering the change in the pressure depending on the position of the breast implant.

[0075] FIG. **9** is a functional block diagram of the apparatus **600, 700** for sensing an abnormality of a breast implant according to some embodiments of the present disclosure.

[0076] As shown in FIG. **9**, the apparatus **600, 700** according to some embodiments of the present disclosure a wireless power transmitting unit **910** for transmitting a power in a wireless manner to the sensor modules **300** of the patches **200, 400** included in the breast implant **110**, an abnormality sensing unit **920** for receiving the pressure signal from at least one of the sensor modules **300** and sensing an abnormality of the breast implant **110** based on the pressure signal, and an alarm generating unit **930** for, upon the abnormality sensing unit **920** sensing the abnormality of the breast implant **110**, generating an alarm to the outside.

[0077] In some embodiments of the present disclosure, the abnormality sensing unit **920** stores, in a memory (not shown), an initial pressure sensed by each of the sensor modules **300** when the breast implant **110** is inserted in the breast **100**, compares pressures indicated by the pressure signals received from the sensor modules **300** with the initial pressures, respectively, and senses the abnormality of the breast implant **110** based on a result of the comparison. For example, when a difference between the pressure indicated the pressure signal received from the sensor module **300** and the initial pressure exceeds a predetermined value (threshold), the abnormality sensing unit **920** determines the abnormality of the breast implant **110**.

[0078] In some embodiments of the present disclosure, upon the abnormality sensing unit **920** sensing the abnormality of the breast implant **110**, the alarm generating unit **930** generates an alarm by means of video or audio signals that can be recognized by the user.

[0079] That is, upon the abnormality sensing unit **920** sensing the abnormality of the breast implant

110, the alarm generating unit **930** visually notifies the user of the abnormality of the breast implant **110** through a display (not shown) and acoustically notifies the user of the abnormality of the breast implant **110** through an alarm sound.

[0080] A breast implant including a sensor module (a pressure sensor) can be manufactured as follows.

[0081] FIGS. **10** to **13** are flowcharts for explaining a method of manufacturing a breast implant according to some embodiments of the present disclosure.

[0082] In the example shown in FIG. **10**, the method of manufacturing the breast implant according to some embodiments of the present disclosure includes a step of forming a shell of the breast implant including an injection hole for injecting a filler (Step **S1010**), a step of injecting the filler of the gel state into the shell through the injection hole (Step **S1020**), and after injecting the filler in the shell, sealing the injection hole with a patch including a first barrier, a second barrier facing the first barrier, and a sensor module between the first barrier and the second barrier, in a manner that the first barrier faces inside the shell and the second barrier faces outside the shell (Step **S1030**).

[0083] In the example shown in FIG. **11**, the method of manufacturing the breast implant according to some embodiments of the present disclosure includes a step of preparing a breast implant including a shell filled with a filler (Step **S1110**) and a step of attaching a patch including a first barrier, a second barrier facing the first barrier, and a sensor module on the outer surface of the first barrier or between the first barrier and the second barrier, in a manner that the first barrier faces the outer surface of the shell (Step **S1120**).

[0084] In the example shown in FIG. **12**, the method of manufacturing the breast implant according to some embodiments of the present disclosure includes a step of forming a shell of the breast implant including an injection hole for injecting a filler (Step **S1210**), a step of injecting the filler of the gel state into the shell through the injection hole (Step **S1220**), after injecting the filler in the shell, sealing the injection hole with a patch including a first barrier, a second barrier facing the first barrier, and a sensor module between the first barrier and the second barrier, in a manner that the first barrier faces inside the shell and the second barrier faces outside the shell (Step **S1230**), and a step of attaching a patch including a sensor module on at least one of the second position and the third position (Step **S1240**).

[0085] In the example shown in FIG. **13**, the method of manufacturing the breast implant according to some embodiments of the present disclosure includes a step of preparing a breast implant including a shell filled with a filler (Step **S1310**), a step of attaching a patch including a first barrier, a second barrier facing the first barrier, and a sensor module on the outer surface of the first barrier or between the first barrier and the second barrier, in a manner that the first barrier faces the outer surface of the shell (Step **S1320**), and a step of attaching a patch including a sensor module on at least one of the second position and the third position (Step **S1330**).

[0086] In the methods of manufacturing a breast implant as described above, the sensor module includes at least one pressure sensor for sensing a pressure and outputting a pressure signal, a controller for transmitting the pressure signal from the at least one pressure sensor, and a wireless power receiving unit for receiving a power in a wireless manner from an external device and providing the power to the controller.

[0087] In the methods of manufacturing a breast implant as described above, the sensor module further includes a battery for storing the power received by the wireless power receiving unit in a wireless manner.

[0088] In this manner, the breast implant according to some embodiments of the present disclosure monitors the pressure applied to the breast implant by means of a sensor tag of a body insertion type including a battery or a sensor tag of a batteryless body insertion type, and when a change of the pressure exceeds a threshold, transmits a signal notifying an abnormality to the apparatus for sensing an abnormality of a breast implant.

[0089] For example, the wireless communication and the wireless power transmission and

reception can be realized by adding a function of the magnetic resonant coupling to an ultra-high frequency (UHF) RFID communication device or using a near field communication (NFC) method. [0090] As described above, according to some embodiments of the present disclosure, it is possible to provide a breast implant including a sensor-equipped patch with which an abnormality of the breast implant can be effectively sensed after inserting the breast implant in a breast.

[0091] Further, according to some embodiments of the present disclosure, it is possible to provide an apparatus for sensing an abnormality of a breast implant, which can effectively sense an abnormality of the breast implant after inserting the breast implant in a breast.

[0092] Moreover, according to some embodiments of the present disclosure, it is possible to provide a method of manufacturing a breast implant including a sensor-equipped patch with which an abnormality of the breast implant can be effectively sensed after inserting the breast implant in a breast.

[0093] The present disclosure should not be limited to these embodiments but various changes and modifications are made by one ordinarily skilled in the art within the subject matter, the spirit and scope of the present disclosure as hereinafter claimed. Specific terms used in this disclosure and drawings are used for illustrative purposes and not to be considered as limitations of the present disclosure. Exemplary embodiments of the present disclosure have been described for the sake of brevity and clarity. Accordingly, one of ordinary skill would understand the scope of the claimed invention is not to be limited by the explicitly described above embodiments but by the claims and equivalents thereof.

Claims

1. A breast implant, comprising: a shell including an injection hole and forming an outer cover of the breast implant; a filler of a gel state injected into the shell through the injection hole; a first patch configured to be attached at a first position and to seal the injection hole, the first patch including a first sensor module; and a second patch configured to be attached on at least one of (i) a second position opposite of the first position and (ii) a third position to which a pressure due to a gravity is applied when the breast implant is inserted in a breast and a body is kept upright without an external pressure applied, the second patch including a second sensor module separate from the first sensor module, wherein the first sensor module is configured to sense a first pressure, and the second sensor module is configured to sense a second pressure different from the first pressure.
2. The breast implant according to claim 1, wherein each of the first patch and the second patch includes: a first barrier forming a first side facing inside the breast implant; a second barrier facing the first barrier and forming a second side opposite to the first side; and a sensor module disposed on the first side or between the first barrier and the second barrier, and wherein each of the first sensor module and the second sensor module includes: at least one pressure sensor configured to sense a pressure and to output a pressure signal; a controller configured to transmit the pressure signal from the at least one pressure sensor; and a wireless power receiving unit configured to receive a power in a wireless manner from an external device and to provide the power to the controller.
3. The breast implant according to claim 1, wherein a diameter of the second patch is smaller than a size of the first patch.
4. The breast implant according to claim 1, wherein when the breast implant is inserted in the breast and the body is kept upright without an external pressure applied: the first sensor module is configured to sense a pressure exerted on the breast implant between the filler and the body, the second sensor module attached on the second position is configured to sense a pressure exerted on the breast implant from outside the body, and the second sensor module attached on the third position is configured to sense a pressure exerted on the breast implant due to the gravity.
5. The breast implant according to claim 1, wherein the second patch is attached on an outer

surface of the shell with the filler injected in a manner that the first barrier makes contact with the outer surface of the shell.

6. An apparatus for sensing an abnormality of a breast implant, the apparatus comprising: a wireless power transmitting unit configured to transmit a power in a wireless manner; an abnormality sensing unit configured to receive a pressure signal from at least one of the first sensor module and the second sensor module of the breast implant according to claim 1 and to sense an abnormality of the breast implant based on the pressure signal; and an alarm generating unit configured to, upon the abnormality sensing unit sensing the abnormality of the breast implant, generate an alarm.

7. The apparatus according to claim 6, wherein the abnormality sensing unit is configured to store, in a memory, an initial pressure sensed by the first sensor module and the second sensor module when the breast implant is inserted in the body, to compare pressures indicated by the pressure signals received from the first sensor module and the second sensor module with the initial pressure, and to sense the abnormality of the breast implant based on a result of comparison.

8. The apparatus according to claim 6, wherein a threshold for determining the abnormality of the breast implant for at least one of the first sensor module and the second sensor module is set to be larger than that of other sensor modules.

9. The apparatus according to claim 6, wherein thresholds for determining the abnormality of the breast implant for the first sensor module and the second sensor module are set to be different from each other.

10. The apparatus according to claim 6, wherein the first sensor module is configured to sense the first pressure exerted on the breast implant between the filler and the body, and the second sensor module is configured to sense the second pressure exerted on the breast implant from either outside of the body or due to gravity.

11. The apparatus according to claim 6, further comprising: a third patch configured to be attached at the third position, the third patch including a third sensor module separate from the first and second sensor modules; wherein the second patch is configured to be attached at the second position opposite of the first position, wherein the second sensor module is configured to sense the second pressure exerted on the breast implant from outside of the body, and wherein the third sensor module is configured to sense a third pressure exerted on the breast implant due to gravity.
